

LI Jin

Tel: +86 18868183767 | Email: Jinli@hdu.edu.cn
GitHub: <https://github.com/Rivflyyy/HappyTorch>

RESEARCH INTERESTS

Multimodal Learning; Large Language Models; Diffusion Models

EDUCATION

Hangzhou Dianzi University

Bachelor of Engineering in Electronic Information Engineering

Hangzhou, China

Sep 2022 - Jun 2026

- **GPA:** 89.59/100
- **Key Courses:** Machine Vision Technology and Application; Signal Detection and Processing; DSP and Intelligent System; Digital Signal Processing; Microelectronics Technology and Testing; EDA Technology
- **Awards:** Bronze Award, 15th Zhejiang Province “Challenge Cup” College Students’ Entrepreneurship Plan Competition (2026); Third Prize, National Undergraduate Physics Experiment Innovation Competition (2025); Third Prize, China College Service Outsourcing Innovation and Entrepreneurship Competition (2025); Zhejiang Provincial Government Scholarship (2025 & 2024); Second Prize, First National ICT Industry-Education Integration Innovation Competition (2024); National Encouragement Scholarship (2023)

PUBLICATIONS

Li, J., Jiang, W., Tian, Y., & Li, Z. (2026). NC-KWS: Few-Shot Class-Incremental Keyword Spotting Based on Neural Collapse. In J. Jia, Z. Wu, L. Gao, G. Huang, & Y. Li (Eds.), *Man-Machine Speech Communication: NCMSC 2025* (pp. 269-276). Springer. https://doi.org/10.1007/978-981-95-5382-2_21

- **Summary:** This study proposes NC-KWS, a few-shot class-incremental keyword spotting framework leveraging Neural Collapse principles. By employing an equiangular tight frame classifier and predefined feature-classifier alignment, it mitigates catastrophic forgetting. Experiments demonstrate that NC-KWS achieves higher accuracy and robustness in later incremental sessions compared to baseline models.

Li, J., Wang, Z., Lu, T., Li, M., Xing, W., & Han, M. (2026). Spectral Logit Sculpting: Adaptive Low-Rank Logit Transformation for Controlled Text Generation. [Accepted for publication in *Proceedings of IEEE International Conference on Acoustics, Speech and Signal Processing*]. arXiv. <https://arxiv.org/abs/2509.25204>

- **Summary:** This study proposes Spectral Logit Sculpting (SLS), an inference-time optimization method that improves LLM reliability by dynamically modulating token distributions via spectral analysis and entropy statistics. Using a sliding buffer and SVD without parameter updates, SLS reduces computational overhead while outperforming baselines in reasoning tasks.

Li, J., Jiang, W., & Hu, J. (2026). KFC: Keyframe Fusion with CTC for User-Defined Keyword Spotting [Accepted for publication in *Proceedings of Interspeech 2026*].

- **Summary:** This study introduces KFC-KWS, a novel keyword spotting framework targeting confusion issues. It utilizes CTC peak posteriors to identify phoneme keyframes for accurate alignment. By integrating audio-text modalities via global-local matching, it significantly surpasses baselines on confusing keyword tasks.

Li, J., Chen, T., Jiang, S., Wang, W., Luo, J., & Wu, C. (2026). OpenVTON-Bench: A Large-Scale High-Resolution Benchmark for Controllable Virtual Try-on Evaluation. arXiv. <https://arxiv.org/abs/2601.22725>

- **Summary:** This study introduces OpenVTON-Bench, a large-scale, high-resolution benchmark for virtual try-on evaluation. It employs DINOv3 clustering and Gemini captioning for dataset construction, and a multi-modal protocol with VLM reasoning and SAM3-based metrics. This approach aligns strongly with human judgment, providing a robust evaluation standard.

Li, J., Hu, W., & Zhou, Z. (2025). MRIE: Enhanced Rainfall Intensity Estimation with Two-Stage Multimodal Deep Learning. *IEEE Access*, 13, 88600-88608. <https://doi.org/10.1109/access.2025.3571440>

- **Summary:** This study proposes MRIE, a two-stage multimodal deep learning framework integrating sensor and audio data for rainfall intensity estimation. Using cross-modal guidance and feature fusion, MRIE significantly reduces MAE and RMSE while increasing R^2 . It demonstrates strong robustness and outperforms existing methods in environmental monitoring.

RESEARCH EXPERIENCE

Research and Implementation of User-Defined Voice Command Recognition Algorithms

Bachelor's Thesis | Awarded as Outstanding Graduation Thesis

Sep 2025 - Present

- **Summary:** This thesis proposes KFC-KWS, a user-defined voice command recognition method using CTC-guided keyframe fusion and multimodal alignment of audio, phoneme, and text. With lightweight random modality masking for robustness, it achieves superior AUC/EER on LibriPhrase and is deployed via a Qt/FastAPI system supporting real-time detection and custom command management.

Scalable Command Recognition System Based on Online Unsupervised Learning

Team Leader | National College Student Innovation and Entrepreneurship Training Program Sep 2024 - Sep 2025

- **Summary:** This research developed a robust edge voice interaction system integrating online adaptation, multimodal fusion, and incremental learning. Achieving 97% accuracy at -5dB SNR with sub-500ms latency on RK3588, it produced 1 patent, 2 papers, and the open HDU-KWS-15k dataset.

Deep Learning-Based Intelligent Fundus Disease Recognition System

Core Member | Zhejiang Province New Seedling Talent Cultivation Program

Feb 2024 - Apr 2025

- **Summary:** This project developed a portable fundus disease screening system using deep learning. It integrates a Raspberry Pi-based hardware platform with a CNN model featuring residual connections, coordinate attention, and spatial-channel reconstruction. Achieving 92.3% lesion segmentation accuracy and 89.7% multi-disease diagnosis accuracy, it includes Grad-CAM visualization and has filed one patent.

Lightweight Keyword Spotting System on Rockchip RK3588 Development Board

Algorithm Design & Software Development | Intelligent Information Processing Lab, HDU Jul 2024 - Jan 2025

- **Summary:** This project developed an end-to-end CTC-based keyword spotting system, trained on fused self-collected and public datasets. The ONNX-optimized model was integrated via C++ prefix beam search decoding and deployed to the edge device. It achieved >95% accuracy under -5-15dB SNR with <200ms latency, ensuring stable on-device performance.

Innovation Patents

- Wang, Y., Yue, K., Li, J., Zhan, Z., & Shi, Z. (2025). **Fundus Image Recognition Method, System and Device Based on Deep Learning** (CN 119444719 A): Developed a portable deep learning system for fundus disease screening using residual connections, coordinate attention, and spatial-channel convolution, achieving efficient remote diagnosis via cloud storage and Grad-CAM visualization
- Jiang, W., Liu, H., Li, J., Tian, Y., & Wu, J. (2025). **Multimodal Multi-language Custom Instruction Recognition Method and System** (CN 120690187 A): Proposed a multimodal framework integrating audio, phoneme, and text features with spike CTC and metric learning, enabling low-latency, customizable voice command recognition across languages without model retraining
- Jiang, W., Li, J., & Liu, H. (2024). **Instruction Recognition Method and System Based on Online Unsupervised Learning** (CN 118366442 A): Designed an online unsupervised learning algorithm for real-time voice command adaptation using entropy minimization and cFSMN-CTC architecture, which reduced latency and improved robustness in dynamic environments without labeled data

INTERNSHIP EXPERIENCE

Hangzhou Renxing Intelligence Technology Co., Ltd.

Hangzhou, China

Algorithm Intern

Nov 2025 - Mar 2026

- Participated in the R&D of ultra-high-resolution controllable image generation algorithms based on Diffusion Models; implemented pixel-space controlled generation pipelines
- Developed a novel frequency-domain Diffusion framework and conducted comprehensive performance validation, reducing FID/LPIPS by 5%-10% compared to baselines
- Optimized training efficiency via DeepSpeed, implementing sequence parallelism and memory optimization strategies, resulting in 30% lower VRAM usage and 1.3x-1.5x higher throughput

Zhejiang University Binjiang Institute

Hangzhou, China

Research Intern

May 2025 - Aug 2025

- Designed reward functions for optimizing multi-turn reasoning in LLMs based on Reinforcement Learning
- Proposed Spectral Logit Sculpting (SLS), an entropy-regulated online adaptive inference optimization method, improving GSM8K accuracy by nearly 100% and Math500 benchmark performance by 2.1%-6.0%
- Conducted research and served as first author on the resulting paper, which was accepted by ICASSP 2026

LANGUAGES & SKILLS

Languages: Chinese (Native); English (Fluent)

Technical Skills: Python (Proficient); C/C++, Java (Familiar); PyTorch, TensorFlow